MATH2400 First Half

Xavier(Xi) Zhang
The University of Queensland

25 Semester 1

Contents

1 Real Numbers **2**
1.1 Bounded . . . 2
1.2 Interval . . . 3
1.3 Absolute value . . . 3

2 Sequence **5**
2.1 Sequences and limits . . . 5
2.2 Continuity of algebraic operations . . . 8
2.3 Limits and inequalities . . . 9
2.4 Monotone sequences . . . 9
2.5 Recursively defined sequences . . . 10
2.6 Subsequences . . . 11
2.7 Bolzano-Weierstrass . . . 12
2.8 Cauchy sequences . . . 12
2.9 Upper and lower limits . . . 14

3 Series **16**
3.1 Basic Properties . . . 16
3.2 Absolute convergence . . . 18
3.3 Tests . . . 19

4 Continuous Functions **21**
4.1 Limits of functions . . . 21
4.2 Continuity . . . 24
4.3 Three Big Results . . . 26
4.4 Uniform continuity . . . 28

5 Derivative **31**
5.1 The Derivative . . . 31
5.2 Mean value theorem . . . 34

Chapter 1

Real Numbers

1.1 Bounded

$$\mathbb{Q} = \left\{ \frac{m}{n} \mid m, n \in \mathbb{Z} \text{ and } n \neq 0 \right\}.$$

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

All of them are ordered.
$\mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ are fields.
$\mathbb{R}$ is a field that is complete and ordered.

Definition 1.1.1: Bounded Above/Below

Let $E \subset S$, where S is an ordered set.

(i) If there exists a $b \in S$ such that $x \leq b$ for all $x \in E$, then we say E is *bounded above* and b is an *upper bound* of E.

(ii) If there exists a $b \in S$ such that $x \geq b$ for all $x \in E$, then we say E is *bounded below* and b is a *lower bound* of E.

(iii) If there exists an upper bound b_0 of E such that whenever b is any upper bound for E we have $b_0 \leq b$, then b_0 is called the *least upper bound* or the *supremum* of E. We write

$$\sup E := b_0.$$

(iv) Similarly, if there exists a lower bound b_0 of E such that whenever b is any lower bound for E we have $b_0 \geq b$, then b_0 is called the *greatest lower bound* or the *infimum* of E. We write

$$\inf E := b_0.$$

$$\sup(\emptyset) = -\infty.$$

If A is not bounded above, then $\sup(A) = \infty$.

$$\inf(\emptyset) = \infty.$$

If A is not bounded below, then $\inf(A) = -\infty$.
$\mathbb{R}^* = \mathbb{R} \cup \{-\infty, \infty\}$
If $\sup A \in A$, then $\max A = \sup A$.
If $\inf A \in A$, then $\min A = \inf A$.

1.2 Interval

$$[a, b] = \{x \mid a \leq x \leq b\} \text{ is closed.}$$

$$[a, b) = \{x \mid a \leq x < b\}$$

$$(a, b] = \{x \mid a < x \leq b\}$$

$$(a, b) = \{x \mid a < x < b\} \text{ is open.}$$

$$[a, \infty] = \{x \mid a \leq x\}$$

$$(-\infty, \infty) = \mathbb{R}$$

Definition 1.2.1: Least-upper-bound property

An ordered set S has the *least-upper-bound property* if every nonempty subset $E \subset S$ that is bounded above has a least upper bound, that is, $\sup E$ exists in S.

Here is some properties in $\mathbb{R}$.

Proposition 1.2.2

Let F be an ordered field and $x, y, z \in F$. Then:

(i) If $x > 0$, then $-x < 0$ (and vice-versa).

(ii) If $x > 0$ and $y < z$, then $xy < xz$.

(iii) If $x < 0$ and $y < z$, then $xy > xz$.

(iv) If $x \neq 0$, then $x^2 > 0$.

(v) If $0 < x < y$, then $0 < \frac{1}{y} < \frac{1}{x}$.

(vi) If $0 < x < y$, then $x^2 < y^2$.

(vii) If $x \leq y$ and $x \leq y$, then $x^2 < y^2$.

1.3 Absolute value

Definition 1.3.1: Absolute value

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Proposition 1.3.2

(i) $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.

(ii) $|-x| = |x|$ for all $x \in \mathbb{R}$.

(iii) $|xy| = |x||y|$ for all $x, y \in \mathbb{R}$.

(iv) $|x|^2 = x^2$ for all $x \in \mathbb{R}$.

(v) $|x| \leq y$ if and only if $-y \leq x \leq y$.

(vi) $-|x| \leq x \leq |x|$ for all $x \in \mathbb{R}$.

Proposition 1.3.3

Triangle Inequality: $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

Proof. We have $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. We add these two inequalities to obtain

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

Then we have $|x + y| \leq |x| + |y|$.

□

Corollary 1.3.4

Let $x, y \in \mathbb{R}$

(i) (reverse triangle inequality) $\big||x| - |y|\big| \leq |x - y|$.

(ii) $|x - y| \leq |x| + |y|$.

Corollary 1.3.5

Let $x_1, x_2, \ldots, x_n \in \mathbb{R}$. *Then*

$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + |x_2| + \cdots + |x_n|.$$

Proof. We proceed by <u>induction</u>. The conclusion holds trivially for $n = 1$, and for $n = 2$ it is the standard triangle inequality. Suppose the corollary holds for n. Take $n + 1$ numbers $x_1, x_2, \ldots, x_n, x_{n+1}$ and first use the standard triangle inequality, then the induction hypothesis

$$|x_1 + x_2 + \cdots + x_n + x_{n+1}| \leq |x_1 + x_2 + \cdots + x_n| + |x_{n+1}|$$

$$\leq |x_1| + |x_2| + \cdots + |x_n| + |x_{n+1}|.$$

□

Chapter 2

Sequence

2.1 Sequences and limits

Definition 2.1.1: Sequence

A *sequence* (of real numbers) is a function $x : \mathbb{N} \to \mathbb{R}$. Instead of $x(n)$ we usually denote the nth element in the sequence by x_n. We use the notation $\{x_n\}$, or more precisely

$$\{x_n\}_{n=1}^{\infty},$$

to denote a sequence.

Example.

The sequence $\{(-1)^n\}$ is the sequence $-1, 1, -1, 1, -1, 1, \ldots$, whereas the set of values, the *range of the sequence*, is just the set $\{-1, 1\}$. We can write this set as $\{(-1)^n : n \in \mathbb{N}\}$. When ambiguity can arise, we use the words *sequence* or *set* to distinguish the two concepts.

Example.

Constant sequence. That is a sequence $\{c\} = c, c, c, c, \ldots$ consisting of a single constant $c \in \mathbb{R}$ repeating indefinitely.

Example.

$\{1/n\}_{n=1}^{\infty}$, or simply $\{1/n\}$, stands for the sequence $1, 1/2, 1/3, 1/4, 1/5, \ldots$. The sequence $\{1/n\}$ is a bounded sequence ($B = 1$ will suffice).

Definition 2.1.2: Convergence of sequence

A sequence $\{x_n\}$ is said to *converge* to a number $x \in \mathbb{R}$, if for every $\varepsilon > 0$, there exists an $M \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq M$. The number x is said to be the *limit* of $\{x_n\}$. We write

$$\lim_{n\to\infty} x_n := x.$$

A sequence that converges is said to be *convergent*. Otherwise, the sequence is said to be *divergent*.

Example.

The constant sequence $c, c, c, c, \ldots$ is convergent and the limit is c. For every $\varepsilon > 0$, we pick $M = 1$.

Claim

The sequence $\{1/n\}$ is convergent and

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Proof for Claim.

Given an $\varepsilon > 0$, we find an $M \in \mathbb{N}$ such that $0 < 1/M < \varepsilon$ (Archimedean property at work). Then for all $n \geq M$ we have that

$$|x_n - 0| = \left|\frac{1}{n}\right| = \frac{1}{n} \leq \frac{1}{M} < \varepsilon.$$

Example.

Let us show the sequence $\left\{\frac{n^2+1}{n^2+n}\right\}$ converges and

$$\lim_{n\to\infty} \frac{n^2+1}{n^2+n} = 1.$$

Given $\varepsilon > 0$, find $M \in \mathbb{N}$ such that $\frac{1}{M+1} < \varepsilon$. Then for any $n \geq M$ we have

$$\begin{aligned}\left|\frac{n^2+1}{n^2+n} - 1\right| &= \left|\frac{n^2+1-(n^2+n)}{n^2+n}\right| = \left|\frac{1-n}{n^2+n}\right| \\ &= \frac{n-1}{n^2+n} \leq \frac{n}{n^2+n} = \frac{1}{n+1} \\ &\leq \frac{1}{M+1} < \varepsilon.\end{aligned}$$

Therefore, $\lim_{n\to\infty} \frac{n^2+1}{n^2+n} = 1$.

Example.

The sequence $\{(-1)^n\}$ is divergent.

Proof: If there were a limit x, then for $\varepsilon = \frac{1}{2}$ we expect an M that satisfies the definition. Suppose such an M exists, then for an even $n \geq M$ we compute

$$\frac{1}{2} > |x_n - x| = |1 - x| \quad \text{and} \quad \frac{1}{2} > |x_{n+1} - x| = |-1 - x|.$$

But

$$2 = |x_n - x_{n+1}| = |1 - x - (-1 - x)| \leq |1 - x| + |-1 - x| < \frac{1}{2} + \frac{1}{2} = 1,$$

and that is a contradiction.

Definition 2.1.3: Bounded sequence

A sequence $\{x_n\}$ is *bounded* if there exists a $B \in \mathbb{R}$ such that

$$|x_n| \leq B \quad \text{for all } n \in \mathbb{N}.$$

In other words, the sequence $\{x_n\}$ is bounded whenever the set $\{x_n : n \in \mathbb{N}\}$ is bounded.

Proposition 2.1.4

A convergent sequence $\{x_n\}$ is bounded.

Proof. Suppose $\{x_n\}$ converges to x. Thus, there exists an $M \in \mathbb{N}$ such that for all $n \geq M$ we have

$$|x_n - x| < 1.$$

Let $B_1 := |x| + 1$ and note that for $n \geq M$ we have

$$\begin{aligned} |x_n| &= |x_n - x + x| \\ &\leq |x_n - x| + |x| \\ &< 1 + |x| = B_1. \end{aligned}$$

The set $\{|x_1|, |x_2|, \ldots, |x_{M-1}|\}$ is a finite set, and hence let

$$B_2 := \max\{|x_1|, |x_2|, \ldots, |x_{M-1}|\}.$$

Let $B := \max\{B_1, B_2\}$. Then for all $n \in \mathbb{N}$, we have

$$|x_n| \leq B.$$

□

We saw

$$\text{convergent} \Rightarrow \text{bounded}$$

$$\text{convergent} \not\Rightarrow \text{bounded}$$

e.g. $\{(-1)^n\}_{n=1}^{\infty}$

Later, we'll see that for monotone sequences:

$$\text{convergent} \Leftrightarrow \text{bounded}$$

Proposition 2.1.5

A convergent sequence has a unique limit.

Proof. Suppose the sequence $\{x_n\}$ has the limit x and the limit y. Take an arbitrary $\varepsilon > 0$. From the definition, find an M_1 such that for all $n \geq M_1$, $|x_n - x| < \varepsilon/2$. Similarly, find an M_2 such that for all $n \geq M_2$ we have $|x_n - y| < \varepsilon/2$. Take $M := \max\{M_1, M_2\}$. For $n \geq M$ (so that both $n \geq M_1$ and $n \geq M_2$) we have

$$\begin{aligned} |y - x| &= |x_n - x - (x_n - y)| \\ &\leq |x_n - x| + |x_n - y| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

As $|y - x| < \varepsilon$ for all $\varepsilon > 0$, then $|y - x| = 0$ and $y = x$. Hence the limit (if it exists) is unique. □

2.2 Continuity of algebraic operations

Proposition 2.2.1

Let $\{x_n\}$ and $\{y_n\}$ be convergent sequences.

(i) *The sequence $\{z_n\}$, where $z_n := x_n + y_n$, converges and*

$$\lim_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n.$$

(ii) *The sequence $\{z_n\}$, where $z_n := x_n - y_n$, converges and*

$$\lim_{n\to\infty}(x_n - y_n) = \lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n - \lim_{n\to\infty} y_n.$$

(iii) *The sequence $\{z_n\}$, where $z_n := x_n y_n$, converges and*

$$\lim_{n\to\infty}(x_n y_n) = \lim_{n\to\infty} z_n = \left(\lim_{n\to\infty} x_n\right)\left(\lim_{n\to\infty} y_n\right).$$

(iv) *If $\lim y_n \neq 0$ and $y_n \neq 0$ for all $n \in \mathbb{N}$, then the sequence $\{z_n\}$, where $z_n := \frac{x_n}{y_n}$, converges and*

$$\lim_{n\to\infty}\frac{x_n}{y_n} = \lim_{n\to\infty} z_n = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n}.$$

Proof. Let us do **(i)**. Suppose $\{x_n\}$ and $\{y_n\}$ are convergent sequences and write $z_n := x_n + y_n$. Let $x := \lim x_n$, $y := \lim y_n$, and $z := x + y$.

Let $\varepsilon > 0$ be given. Find an M_1 such that for all $n \geq M_1$ we have $|x_n - x| < \varepsilon/2$. Find an M_2 such that for all $n \geq M_2$ we have $|y_n - y| < \varepsilon/2$. Take $M := \max\{M_1, M_2\}$. For all $n \geq M$ we have

$$\begin{aligned}|z_n - z| &= |(x_n + y_n) - (x + y)| = |x_n - x + y_n - y| \\ &\leq |x_n - x| + |y_n - y| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

Therefore (i) is proved.

□

Proposition 2.2.2

Let $\{x_n\}$ be a convergent sequence such that $x_n \geq 0$. Then

$$\lim_{n\to\infty}\sqrt{x_n} = \sqrt{\lim_{n\to\infty} x_n}.$$

Proposition 2.2.3

If $\{x_n\}$ is a convergent sequence, then $\{|x_n|\}$ is convergent and

$$\lim_{n\to\infty}|x_n| = \left|\lim_{n\to\infty} x_n\right|.$$

Proof. Set $x = \lim_{n\to\infty} x_n$.

$$\forall\varepsilon > 0,\ \exists M \text{ such that } \forall n \geq M,\ |x_n - x| < \varepsilon.$$

We simply note the reverse triangle inequality

$$||x_n| - |x|| \leq |x_n - x|.$$

Hence, if $|x_n - x|$ can be made arbitrarily small, so can $||x_n| - |x||$.

□

2.3 Limits and inequalities

Lemma 2.3.1: Squeeze lemma

Let $\{a_n\}, \{b_n\}$, and $\{x_n\}$ be sequences such that

$$a_n \leq x_n \leq b_n \quad \text{for all } n \in \mathbb{N}.$$

Suppose $\{a_n\}$ and $\{b_n\}$ converge and

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n.$$

Then $\{x_n\}$ converges and

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n.$$

Proof. For $\varepsilon > 0$,

$$\exists M_1, \forall n \geq M_1, \quad |a_n - L| < \varepsilon$$
$$\exists M_2, \forall n \geq M_2, \quad |b_n - L| < \varepsilon$$

Set $M = \max\{M_1, M_2\}$, so for $n \geq M$:

$$L - \varepsilon \leq a_n \leq x_n \leq b_n \leq L + \varepsilon$$

$$L - \varepsilon < x_n < L + \varepsilon$$

$$|x_n - L| < \varepsilon.$$

□

2.4 Monotone sequences

Definition 2.4.1

A sequence $\{x_n\}$ is *monotone increasing* if $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$. A sequence $\{x_n\}$ is *monotone decreasing* if $x_n \geq x_{n+1}$ for all $n \in \mathbb{N}$. If a sequence is either monotone increasing or monotone decreasing, we can simply say the sequence is *monotone*. Some authors also use the word *monotonic*.

Theorem 2.4.2

A monotone sequence $\{x_n\}$ is bounded if and only if it is convergent. *Furthermore, if $\{x_n\}$ is monotone increasing and bounded, then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

If $\{x_n\}$ is monotone decreasing and bounded, then

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

We need some theorems to prove this.

Proposition 2.4.3

If $S \subset \mathbb{R}$ is a nonempty set, bounded from above, then for every $\varepsilon > 0$ there exists $x \in S$ such that $(\sup S) - \varepsilon < x \leq \sup S$.

Theorem 2.4.4: Real Number

There exists a unique **ordered field** containing $\mathbb{Q}$ with the **least upper bound property**. We denote this field by $\mathbb{R}$.

Proof. Let us suppose the sequence is monotone increasing. Suppose the sequence is bounded, so there exists a B such that $x_n \leq B$ for all n, that is, the set $\{x_n : n \in \mathbb{N}\}$ is bounded from above. Let

$$x := \sup\{x_n : n \in \mathbb{N}\}.$$

Let $\varepsilon > 0$ be arbitrary. As x is the supremum, then there must be at least one $M \in \mathbb{N}$ such that $x_M > x - \varepsilon$ (because x is the supremum). As $\{x_n\}$ is monotone increasing, there $\exists M \in \mathbb{N}$ such that $x - \varepsilon < x_M$. Then,

$$x - \varepsilon < x_M \leq x_{M+1} \leq x_{M+2} \leq \dots$$

So, $\forall n \geq M$

$$x - \varepsilon < x_n$$

$$x - x_n < \varepsilon$$

Hence by $x_n < x$,

$$|x_n - x| = x - x_n \leq \varepsilon.$$

Therefore, the sequence converges to x. We already know that a convergent sequence is bounded, which completes the other direction of the implication.

The proof for monotone decreasing sequences is similar.

□

2.5 Recursively defined sequences

Example.

Let $\{x_n\}$ be defined by $x_1 := 2$ and

$$x_{n+1} := x_n - \frac{x_n^2 - 2}{2x_n}.$$

Proof. We must first find out if this sequence is well defined; we must show we never divide by zero. Then we must find out if the sequence converges. Only then can we attempt to find the limit.

First let us prove x_n exists and $x_n > 0$ for all n (so the sequence is well defined and bounded below). Let us show this by *induction*. We know that $x_1 = 2 > 0$. For the induction step, suppose $x_n > 0$. Then

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{2x_n^2 - x_n^2 + 2}{2x_n} = \frac{x_n^2 + 2}{2x_n}.$$

If $x_n > 0$, then $x_n^2 + 2 > 0$ and hence $x_{n+1} > 0$.

Next let us show that the sequence is monotone decreasing. If we show that $x_n^2 - 2 \geq 0$ for all n, then $x_{n+1} \leq x_n$ for all n. Obviously, $x_1^2 - 2 = 4 - 2 = 2 > 0$. For an arbitrary n, we have

$$x_{n+1}^2 - 2 = \left(\frac{x_n^2 + 2}{2x_n}\right)^2 - 2 = \frac{x_n^4 + 4x_n^2 + 4 - 8x_n^2}{4x_n^2} = \frac{(x_n^2 - 2)^2}{4x_n^2}.$$

Since any number squared is nonnegative, we have that $x_{n+1}^2 - 2 \geq 0$ for all n. Therefore, $\{x_n\}$ is monotone decreasing and bounded ($x_n > 0$ for all n), and the limit exists. It remains to find the limit.

Let us write

$$2x_n x_{n+1} = x_n^2 + 2.$$

Since $\{x_n\}$ is the 1-tail of $\{x_n\}$, it converges to the same limit. Let us define $x := \lim x_n$. We take the limit of both sides to obtain

$$2x^2 = x^2 + 2.$$

or $x^2 = 2$. As $x_n > 0$ for all n, we get $x \geq 0$, and therefore $x = \sqrt{2}$.

□

2.6 Subsequences

Definition 2.6.1: Subsequences

Let $\{x_n\}$ be a sequence. Let $\{n_i\}$ be a strictly increasing sequence of natural numbers (that is $n_1 < n_2 < n_3 < \cdots$). The sequence

$$\{x_{n_i}\}_{i=1}^{\infty}$$

is called a *subsequence* of $\{x_n\}$.

Example.

For example, take the sequence $\{1/n\}$. The sequence $\{1/3n\}$ is a subsequence. To see how these two sequences fit in the definition, take $n_i := 3i$.

$$n_i = 3, 6, 9, ...$$

$$\{1/3n\} = 1/3, 1/6, 1/9, ...$$

Example.

For example, take the sequence $\{x_n\}_{n=1}^{\infty} = \{\sin \frac{n\pi}{2}\}_{n=1}^{\infty}$, which has $T = 4$. Take $n_i := 4i - 3$, which is $1, 5, 9, ...$

$$\{x_{n_i}\}_{i=1}^{\infty} = \{1\}_{n=1}^{\infty}$$

Lemma 2.6.2

For a strictly increasing sequences of $\mathbb{N}$, $n_1 < n_2 < n_3 < ...$, $n_i \geq i$ for all $i \in \mathbb{N}$.

Proof. Base case: $n_1 \geq 1$. Assume $n_i \geq i$ and show $n_{i+1} \geq i + 1$. Since $n_{i+1} \geq n_i$, we know $n_{i+1} \geq n_i \geq i \Rightarrow n_{i+1} \geq i \Rightarrow n_{i+1} > i + 1$

□

Proposition 2.6.3

If $\{x_n\}$ is a convergent sequence, then any subsequence $\{x_{n_i}\}$ is also convergent and

$$\lim_{n\to\infty} x_n = \lim_{i\to\infty} x_{n_i}.$$

Proof. Suppose $\lim_{n\to\infty} x_n = x$. That means that for every $\epsilon > 0$ we have an $M \in \mathbb{N}$ such that for all $n \geq M$

$$|x_n - x| < \epsilon.$$

We know that $n_i \geq i$. Hence $i \geq M$ implies $n_i \geq M$. Thus, for all $i \geq M$ we have

$$|x_{n_i} - x| < \epsilon,$$

and we are done. □

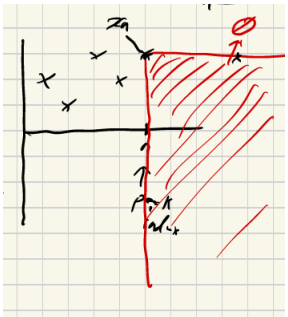
Lemma 2.6.4: Peak Lemma

Every sequence $(x_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}}$ has a monotone subsequence.

Proof. Take $x_1, x_2, x_3, ...$, call index n a peak-index if $x_n \geq x_k$ $\forall k \geq n$, call x_n a peak.
There can be two cases:
Case I: (infinite many peak indices)
n_1: index of the first, n_2: index of the second, ...

$$n_1 < n_2 < n_3 < ... \Rightarrow x_{n_1} \geq x_{n_2} \geq x_{n_3}$$

Hence $\{x_{n_i}\}_{i=1}^{\infty}$ is monotone increasing subsequence.
Case II: (finite number of peak indices)
Let N be the largest peak number. Then $\forall n > N$, x_n is not a peak.
Take $n_1 = N + 1$. Take n_2, $\exists n_2 > n_1$ with $x_{n_2} > x_{n_1}$. Take n_3, $\exists n_3 > n_2$ with $x_{n_3} > x_{n_2}$.
Our subsequence indices is $n_1 < n_2 < n_3 < ....$ Then we have $x_{n_1} \geq x_{n_2} \geq x_{n_3}$, which is monotone increasing. □



2.7 Bolzano-Weierstrass

Theorem 2.7.1: Bolzano-Weierstrass

Suppose a sequence $\{x_n\}$ of real numbers is bounded. Then there exists a convergent subsequence $\{x_{n_j}\}$.

Proof. The sequence $\{x_n\}_{n=1}^{\infty}$ has a monotone subsequence $\{x_{n_i}\}_{i=1}^{\infty}$ by the peak lemma.
Since $\{x_n\}_{n=1}^{\infty}$ is bounded, $\{x_{n_i}\}_{i=1}^{\infty}$ is bounded.
By the monotone convergence theorem, $\{x_{n_i}\}_{i=1}^{\infty}$ converges. □

2.8 Cauchy sequences

Definition 2.8.1

A sequence $\{x_n\}$ is a *Cauchy sequence** if for every $\varepsilon > 0$ there exists an $M \in \mathbb{N}$ such that for all $n \geq M$ and all $k \geq M$ we have

$$|x_n - x_k| < \varepsilon.$$

Example.

The sequence $\{1/n\}$ is a Cauchy sequence.

Proof. Given $\varepsilon > 0$, find M such that $M > 2/\varepsilon$. Then for $n, k \geq M$ we have that $1/n < \varepsilon/2$ and $1/k < \varepsilon/2$. Therefore for $n, k \geq M$ we have

$$\left|\frac{1}{n} - \frac{1}{k}\right| \leq \left|\frac{1}{n}\right| + \left|\frac{1}{k}\right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Example.

The sequence $\{(-1)^n\}_{n=1}^{\infty}$ is not a Cauchy sequence.

Proof. For any $M \in \mathbb{N}$, take $k \geq M$ as an *even* number and let $l = k + 1$ (*odd*).

$$\left|(-1)^k - (-1)^l\right| = |1 - (-1)| = 2 \not< \varepsilon$$

for $\varepsilon \leq 2$. □

Proposition 2.8.2

A Cauchy sequence is bounded.

Proof. Suppose $\{x_n\}$ is Cauchy. Pick M such that for all $n, k \geq M$ we have $|x_n - x_k| < 1$. In particular, we have that for all $n \geq M$

$$|x_n - x_M| < 1.$$

Or by the reverse triangle inequality, $|x_n| - |x_M| \leq |x_n - x_M| < 1$. Hence for $n \geq M$ we have

$$|x_n| < 1 + |x_M|.$$

Let

$$B := \max\{|x_1|, |x_2|, \ldots, |x_{M-1}|, 1 + |x_M|\}.$$

Then $|x_n| \leq B$ for all $n \in \mathbb{N}$.

□

Theorem 2.8.3

A sequence of real numbers is Cauchy if and only if it converges.

Proof. Let $\varepsilon > 0$ be given and suppose $\{x_n\}$ converges to x. Then there exists an M such that for $n \geq M$ we have

$$|x_n - x| < \frac{\varepsilon}{2}.$$

Hence for $n \geq M$ and $k \geq M$ we have

$$|x_n - x_k| = |x_n - x + x - x_k| \leq |x_n - x| + |x_k - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Alright, that direction was easy. Now assume $(x_n)_{n=1}^{\infty}$ is a Cauchy sequence. Hence, it is bounded. Thus, there exists a subsequence $(x_{n_i})_{i=1}^{\infty}$ that converges to some limit L. Let $\varepsilon > 0$. Since (x_{n_i}) converges, there exists M_2 such that:

$$\forall i \geq M_2, \quad |x_{n_i} - L| \leq \frac{\varepsilon}{2}$$

Since (x_n) is a Cauchy sequence, there exists M_1 such that for all $k, l \geq M_1$:

$$|x_k - x_l| \leq \frac{\varepsilon}{2}$$

Take $M = \max\{M_1, M_2\}$. Consider $x_{n_{M+1}}$. Now, for all $n \geq M$:

$$\begin{aligned} |x_n - L| &= |x_n - x_{n_{M+1}} + x_{n_{M+1}} - L| \\ &\leq |x_n - x_{n_{M+1}}| + |x_{n_{M+1}} - L| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

□

2.9 Upper and lower limits

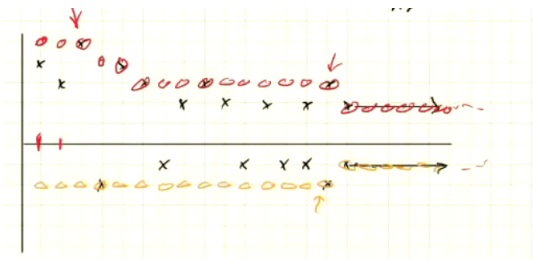
Definition 2.9.1

Let $\{x_n\}$ be a bounded sequence. Let

$$a_n := \sup\{x_k : k \geq n\} \quad \text{and} \quad b_n := \inf\{x_k : k \geq n\}.$$

Define

$$\begin{aligned} \limsup_{n\to\infty} x_n &:= \lim_{n\to\infty} a_n, \\ \liminf_{n\to\infty} x_n &:= \lim_{n\to\infty} b_n. \end{aligned}$$



Example.

Let $\{x_n\}$ be defined by

$$x_n := \begin{cases} \frac{n+1}{n} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases}$$

Let us compute the lim inf and lim sup of this sequence. First the limit inferior:

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} (\inf\{x_k : k \geq n\}) = \lim_{n\to\infty} 0 = 0.$$

For the limit superior we write

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} (\sup\{x_k : k \geq n\}).$$

It is not hard to see that

$$\sup\{x_k : k \geq n\} = \begin{cases} \frac{n+1}{n} & \text{if } n \text{ is odd,} \\ \frac{n+2}{n+1} & \text{if } n \text{ is even.} \end{cases}$$

We leave it to the reader to show that the limit is 1. That is,

$$\limsup_{n\to\infty} x_n = 1.$$

Do note that the sequence $\{x_n\}$ is not a convergent sequence.

Proposition 2.9.2

Let $\{x_n\}$ be a bounded sequence. Let a_n and b_n be as in the definition above.

(i) *The sequence $\{a_n\}$ is bounded monotone decreasing and $\{b_n\}$ is bounded monotone increasing. In particular,* $\liminf x_n$ *and* $\limsup x_n$ *exist.*

(ii) $\limsup_{n\to\infty} x_n = \inf\{a_n : n \in \mathbb{N}\}$ and $\liminf_{n\to\infty} x_n = \sup\{b_n : n \in \mathbb{N}\}$.

(iii) $\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n$.

Proof. Let us see why $\{a_n\}$ is a decreasing sequence. As a_n is the least upper bound for $\{x_k : k \geq n\}$, it is also an upper bound for the subset $\{x_k : k \geq (n+1)\}$. Therefore, a_{n+1}, the least upper bound for $\{x_k : k \geq (n+1)\}$, has to be less than or equal to a_n, that is, $a_n \geq a_{n+1}$. Similarly (an exercise), $\{b_n\}$ is an increasing sequence. It is left as an exercise to show that if $\{x_n\}$ is bounded, then a_n and b_n must be bounded.

The second item in the proposition follows as the sequences $\{a_n\}$ and $\{b_n\}$ are monotone.

For the third item, we note that $b_n \leq a_n$, as the inf of a set is less than or equal to its sup. We know that $\{a_n\}$ and $\{b_n\}$ converge to the lim sup and the lim inf (respectively). We apply *Lemma 2.2.3* to obtain

$$\lim_{n\to\infty} b_n \leq \lim_{n\to\infty} a_n.$$

□

Theorem 2.9.3

If $\{x_n\}$ is a bounded sequence, then there exists a subsequence $\{x_{n_k}\}$ such that

$$\lim_{k\to\infty} x_{n_k} = \limsup_{n\to\infty} x_n.$$

Similarly, there exists a (perhaps different) subsequence $\{x_{m_k}\}$ such that

$$\lim_{k\to\infty} x_{m_k} = \liminf_{n\to\infty} x_n.$$

Theorem 2.9.4

Let $\{x_n\}$ be a bounded sequence. Then $\{x_n\}$ converges if and only if

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n.$$

Furthermore, if $\{x_n\}$ converges, then

$$\lim_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n.$$

Chapter 3

Series

3.1 Basic Properties

Definition 3.1.1: Series

Given a sequence $\{x_n\}$, we write the formal object

$$\sum_{n=1}^{\infty} x_n \quad \text{or sometimes just} \quad \sum x_n$$

and call it a *series*. A series *converges*, if the sequence $\{s_k\}$ defined by

$$s_k := \sum_{n=1}^{k} x_n = x_1 + x_2 + \cdots + x_k,$$

converges. The numbers s_k are called *partial sums*. If $x := \lim s_k$, we write

$$\sum_{n=1}^{\infty} x_n = x.$$

In this case, we cheat a little and treat $\sum_{n=1}^{\infty} x_n$ as a number.

On the other hand, if the sequence $\{s_k\}$ diverges, we say the series is *divergent*. In this case, $\sum x_n$ is simply a formal object and not a number.

In other words, for a convergent series we have

$$\sum_{n=1}^{\infty} x_n = \lim_{k\to\infty} S_k = \lim_{k\to\infty} \sum_{n=1}^{k} x_n.$$

Proposition 3.1.2

For $-1 < r < 1$, the *geometric series*

$$\sum_{n=0}^{\infty} r^n$$

converges. In fact, $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$.

Proof. Using induction, we prove

$$\sum_{n=0}^{k} r^n = \frac{1 - r^{k+1}}{1 - r}, \quad \forall r \neq 1$$

$$s_k = \frac{1 - r^{k+1}}{1 - r}$$

Thus,

$$\lim_{k \to \infty} \frac{1 - r^{k+1}}{1 - r} = \frac{1}{1 - r}$$

□

Definition 3.1.3

A series $\sum x_n$ is said to be *Cauchy* or a *Cauchy series*, if the sequence of partial sums $\{s_n\}$ is a Cauchy sequence.

A sequence of real numbers converges if and only if it is Cauchy. Therefore a series is convergent if and only if it is Cauchy.

Definition 3.1.4

The series $\sum x_n$ is Cauchy if for every $\varepsilon > 0$, there exists an $M \in \mathbb{N}$ such that for every $n \geq M$ and every $k > n$ we have

$$\left| \sum_{j=n+1}^{k} x_j \right| < \varepsilon.$$

Proof.

$$|S_k - S_n| < \epsilon$$

□

Proposition 3.1.5

Let $\sum x_n$ be a convergent series. Then the sequence $\{x_n\}$ is convergent and

$$\lim_{n \to \infty} x_n = 0.$$

Proof. Let $\varepsilon > 0$ be given. As $\sum x_n$ is convergent, it is Cauchy. Thus we find an M such that for every $n \geq M$ we have

$$\varepsilon > \left| \sum_{j=n+1}^{n+1} x_j \right| = |x_{n+1}|.$$

Hence for every $n \geq M + 1$ we have $|x_n - 0| < \varepsilon$. □

Claim

$|r^n|$ diverges if $|r| \geq 1$

Proof for Claim.

Take $\sum_{n=1}^{\infty} r^n$ with $1 \leq |r|$.

But r^n cannot have a limit 0:

$$|r^n - 0| = |r^n| = |r|^n \geq 1^n = 1.$$

Claim

There exists a series $\sum_{n=1}^{\infty} x_n$ with

$$\lim_{n\to\infty} x_n = 0,$$

that diverges.

Proof for Claim.

Consider the *harmonic series*

$$\sum_{n=1}^{\infty} \frac{1}{n},$$

which diverges.

We will show that the sequence of partial sums is unbounded, and hence cannot converge. Write the partial sums s_n for $n = 2^k$ as:

$$\begin{aligned}
s_1 &= 1, \\
s_2 &= (1) + \left(\frac{1}{2}\right), \\
s_4 &= (1) + \left(\frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right), \\
s_8 &= (1) + \left(\frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right), \\
&\vdots \\
s_{2^k} &= 1 + \sum_{j=1}^{k} \left(\sum_{m=2^{j-1}+1}^{2^j} \frac{1}{m} \right).
\end{aligned}$$

We note that $\frac{1}{3} + \frac{1}{4} \geq \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \geq \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$. More generally,

$$\sum_{m=2^{k-1}+1}^{2^k} \frac{1}{m} \geq \sum_{m=2^{k-1}+1}^{2^k} \frac{1}{2^k} = (2^k - 1)\frac{1}{2^k} = \frac{1}{2}.$$

Therefore,

$$s_{2^k} = 1 + \sum_{j=1}^{k} \left(\sum_{m=2^{k-1}+1}^{2^k} \frac{1}{m} \right) \geq 1 + \sum_{j=1}^{k} \frac{1}{2} = 1 + \frac{k}{2}.$$

As $\left\{\frac{k}{2}\right\}$ is unbounded by the **Archimedean property**, that means that $\{s_{2^k}\}$ is unbounded, and therefore $\{s_n\}$ is unbounded. Hence $\{s_n\}$ diverges, and consequently $\sum \frac{1}{n}$ diverges.

3.2 Absolute convergence

Proposition 3.2.1

If $x_n \geq 0$ for all n, then $\sum x_n$ converges if and only if the sequence of partial sums is bounded from above.

Proof.

$$x_n \geq 0 \Rightarrow S_k \text{ is monotone increasing and bounded} \Rightarrow S_k \text{ is convergent}$$

□

Definition 3.2.2

A series $\sum x_n$ *converges absolutely* if the series $\sum |x_n|$ converges. If a series converges, but does not converge absolutely, we say it is *conditionally convergent*.

Proposition 3.2.3

If the series $\sum x_n$ converges absolutely, then it converges.

Proof. A series is convergent if and only if it is Cauchy. Hence suppose $\sum |x_n|$ is Cauchy. That is, for every $\varepsilon > 0$, there exists an M such that for all $k \geq M$ and $n > k$ we have

$$\sum_{j=k+1}^{n} |x_j| = \left| \sum_{j=k+1}^{n} |x_j| \right| < \varepsilon.$$

We apply the triangle inequality for a finite sum to obtain

$$\left| \sum_{j=k+1}^{n} x_j \right| \leq \sum_{j=k+1}^{n} |x_j| < \varepsilon.$$

Hence $\sum x_n$ is Cauchy and therefore it converges. □

Example.

The series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges.

But does not converge absolutely:

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which diverges.

Hence, it is *conditionally convergent*.

3.3 Tests

Proposition 3.3.1

(Comparison test). *Let $\sum x_n$ and $\sum y_n$ be series such that $0 \leq x_n \leq y_n$ for all $n \in \mathbb{N}$.*

(i) If $\sum y_n$ converges, then so does $\sum x_n$.

(ii) If $\sum x_n$ diverges, then so does $\sum y_n$.

Proof. Since the terms of the series are all nonnegative, the sequences of partial sums are both monotone increasing. Since $x_n \leq y_n$ for all n, the partial sums satisfy for all k

$$\sum_{n=1}^{k} x_n \leq \sum_{n=1}^{k} y_n. \tag{2.1}$$

If the series $\sum y_n$ converges, the partial sums for the series are bounded. Therefore, the right-hand side of (2.1) is bounded for all k. Hence the partial sums for $\sum x_n$ are also bounded. Since the partial sums are a monotone increasing sequence, they are convergent. The first item is thus proved.

□

Proposition 3.3.2

(*p-series or the p-test*). *For $p \in \mathbb{R}$, the series*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Proposition 3.3.3

(Ratio test). *Let $\sum x_n$ be a series such that*

$$L := \lim_{n\to\infty} \frac{|x_{n+1}|}{|x_n|}$$

exists. Then

(i) If $L < 1$, then $\sum x_n$ converges absolutely.

(ii) If $L > 1$, then $\sum x_n$ diverges.

Proposition 3.3.4

(Root test). *Let $\sum x_n$ be a series and let*

$$L := \limsup_{n\to\infty} |x_n|^{1/n}.$$

Then

(i) If $L < 1$, then $\sum x_n$ converges absolutely.

(ii) If $L > 1$, then $\sum x_n$ diverges.

Proposition 3.3.5

(Alternating series). *Let $\{x_n\}$ be a monotone decreasing sequence of positive real numbers such that $\lim x_n = 0$. Then*

$$\sum_{n=1}^{\infty} (-1)^n x_n$$

converges.

Chapter 4

Continuous Functions

4.1 Limits of functions

Definition 4.1.1

Let $f : \mathbb{R} \to \mathbb{R}$ be a function and $c \in \mathbb{R}$. Suppose there exists an $L \in \mathbb{R}$ and for every $\varepsilon > 0$, there exists a $\delta > 0$ such that whenever $x \in \mathbb{R} \setminus \{c\}$ and $0 < |x - c| < \delta$, then

$$|f(x) - L| < \varepsilon.$$

In this case we say $f(x)$ *converges* to L as x goes to c. We say L is the *limit* of $f(x)$ as x goes to c. We write

$$\lim_{x \to c} f(x) := L,$$

or

$$f(x) \to L \quad \text{as} \quad x \to c.$$

If no such L exists, then we say that the limit does not exist or that f *diverges* at $c \Leftrightarrow \exists \epsilon > 0, \forall \delta > 0$ such that $\exists x : 0 < |x - c| < \delta$, then $|f(x) - L| \geq \varepsilon$.

Example.

Show that

$$\lim_{x \to 3} \frac{x-1}{2} = 1$$

Proof. Let $\epsilon > 0$ be given.
Set $\delta = \epsilon$. Hence, for all x, if $0 < |x - 3| < \delta$, then:

$$\left| \frac{x-1}{2} - 1 \right| = \left| \frac{x-3}{2} \right| < \frac{\epsilon}{2} < \epsilon$$

□

Example.

Show that

$$\lim_{x \to c} (mx + c) = mc + b, m, c \in \mathbb{R}$$

Proof. Let $\epsilon > 0$ be given.
Set $\delta = \frac{\epsilon}{|m|}$. Hence, for all x, if $0 < |x - c| < \delta$, then:

$$|mx + c - (mc + b)| = |m| \, |x - c| < \epsilon$$

□

Example.

Show that

$$\lim_{x \to 4} x^2 = 16$$

Proof. Let $\epsilon > 0$ be given.

Set $\delta = \min\{\frac{\epsilon}{9}, 1\}$. Hence, for all x, if $0 < |x - 4| < \delta$, then:

$$\left|x^2 - 16\right| = |(x+4)(x-4)| \leq |x-4|\,(|x| + |4|) \leq |x-4|\,9 < \epsilon$$

Since $|x| - |4| \leq |x - 4| < 1 \Rightarrow |x| + |4| < 9$ □

Proposition 4.1.2

The limit of a function at a point is unique if it exists.

Proof. Let L_1 and L_2 be two numbers that both satisfy the definition. Take an $\epsilon > 0$ and find a $\delta_1 > 0$ such that $|f(x) - L_1| < \epsilon/2$ for all $x \in S \setminus \{c\}$ with $|x - c| < \delta_1$. Also find $\delta_2 > 0$ such that $|f(x) - L_2| < \epsilon/2$ for all $x \in S \setminus \{c\}$ with $|x - c| < \delta_2$. Put $\delta := \min\{\delta_1, \delta_2\}$. Suppose $x \in S$, $|x - c| < \delta$, and $x \neq c$. Then

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |L_1 - f(x)| + |f(x) - L_2| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

As $|L_1 - L_2| < \epsilon$ for arbitrary $\epsilon > 0$, then $L_1 = L_2$.

□

Definition 4.1.3: Sequential limit

Let $S \subset \mathbb{R}$ and $c \in S$. Let $f : S \to \mathbb{R}$ be a function. Then $f(x) \to L$ as $x \to c$, if and only if for every sequence $\{x_n\}$ of numbers such that $x_n \in S \setminus \{c\}$ for all n, and such that $\lim x_n = c$, we have that the sequence $\{f(x_n)\}$ converges to L.

Example.

Show that

$$\lim_{x \to 3} \frac{x-1}{2} = 1$$

Proof. Let $(x_n)_{n=1}^{n}$ be a sequence with $\lim_{n\to\infty} x_n = 3, \lim_{n\to\infty}(x_n - 1) = 3 - 1 = 2, \lim_{n\to\infty} \frac{(x_n - 1)}{2} = 1$ □

Proposition 4.1.4

Sequential limit $\Leftrightarrow \epsilon, \delta$ limit

Proof. Suppose $f(x) \to L$ as $x \to c$, and $\{x_n\}$ is a sequence such that $x_n \in S \setminus \{c\}$ and $\lim x_n = c$. We wish to show that $\{f(x_n)\}$ converges to L. Let $\varepsilon > 0$ be given. Find a $\delta > 0$ such that if $x \in S \setminus \{c\}$ and $|x - c| < \delta$, then $|f(x) - L| < \varepsilon$. As $\{x_n\}$ converges to c, find an M such that for $n \geq M$ we have that $|x_n - c| < \delta$. Therefore, for $n \geq M$,

$$|f(x_n) - L| < \varepsilon.$$

Thus $\{f(x_n)\}$ converges to L.

For the other direction, we use proof by contrapositive. Suppose it is not true that $f(x) \to L$ as $x \to c$. The negation of the definition is that there exists an $\varepsilon > 0$ such that for every $\delta > 0$ there exists an $x \in S \setminus \{c\}$, where $|x - c| < \delta$ and $|f(x) - L| \geq \varepsilon$.

Let us use $1/n$ for δ in the above statement to construct a sequence $\{x_n\}$. We have that there exists an $\varepsilon > 0$ such that for every n, there exists a point $x_n \in S \setminus \{c\}$, where $|x_n - c| < 1/n$ and $|f(x_n) - L| \geq \varepsilon$. The sequence $\{x_n\}$ just constructed converges to c, but the sequence $\{f(x_n)\}$ does not converge to L. And we are done. □

Example.

$$\lim_{x\to 0} \sin(1/x) \text{ does not exist, but } \lim_{x\to 0} x\sin(1/x) = 0.$$

Proof. Let us work with $\sin(1/x)$ first. Let us define the sequence $x_n := \frac{1}{\pi n + \pi/2}$. It is not hard to see that $\lim x_n = 0$. Furthermore,

$$\sin(1/x_n) = \sin(\pi n + \pi/2) = (-1)^n.$$

Therefore, $\{\sin(1/x_n)\}$ does not converge. Thus, $\lim_{x\to 0} \sin(1/x)$ does not exist.

Now let us look at $x\sin(1/x)$. Let x_n be a sequence such that $x_n \neq 0$ for all n and such that $\lim x_n = 0$. Notice that $|\sin(t)| \leq 1$ for any $t \in \mathbb{R}$. Therefore,

$$|x_n \sin(1/x_n) - 0| = |x_n||\sin(1/x_n)| \leq |x_n|.$$

As x_n goes to 0, then $|x_n|$ goes to zero, and hence $\{x_n \sin(1/x_n)\}$ converges to zero. Thus, $\lim_{x\to 0} x\sin(1/x) = 0$. □

Corollary 4.1.5

Let $S \subset \mathbb{R}$ and c be a point of S. Let $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ be functions. Suppose the limits of $f(x)$ and $g(x)$ as x goes to c both exist, and that

$$f(x) \leq g(x) \quad \text{for all } x \in S.$$

Then

$$\lim_{x\to c} f(x) \leq \lim_{x\to c} g(x).$$

Lemma 4.1.6

Let $\{x_n\}$ and $\{y_n\}$ be convergent sequences and

$$x_n \leq y_n,$$

for all $n \in \mathbb{N}$. Then

$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n.$$

Proof. Take $\{x_n\}$ be a sequence of numbers in $S \setminus \{c\}$ that converges to c. Let

$$L_1 := \lim_{x\to c} f(x), \quad \text{and} \quad L_2 := \lim_{x\to c} g(x).$$

We know $\{f(x_n)\}$ converges to L_1 and $\{g(x_n)\}$ converges to L_2. We also have $f(x_n) \leq g(x_n)$. We obtain $L_1 \leq L_2$ using Lemma above.

□

Corollary 4.1.7

Let $S \subset \mathbb{R}$ and c be a cluster point of S. Let $f : S \to \mathbb{R}$, $g : S \to \mathbb{R}$, and $h : S \to \mathbb{R}$ be functions. Suppose

$$f(x) \leq g(x) \leq h(x) \quad \text{for all } x \in S,$$

and the limits of $f(x)$ and $h(x)$ as x goes to c both exist, and

$$\lim_{x \to c} f(x) = \lim_{x \to c} h(x).$$

Then the limit of $g(x)$ as x goes to c exists and

$$\lim_{x \to c} g(x) = \lim_{x \to c} f(x) = \lim_{x \to c} h(x).$$

Corollary 4.1.8

Let $S \subset \mathbb{R}$ and c be a cluster point of S. Let $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ be functions. Suppose limits of $f(x)$ and $g(x)$ as x goes to c both exist. Then

(i) $\lim_{x \to c}(f(x) + g(x)) = \left(\lim_{x \to c} f(x)\right) + \left(\lim_{x \to c} g(x)\right).$

(ii) $\lim_{x \to c}(f(x) - g(x)) = \left(\lim_{x \to c} f(x)\right) - \lim_{x \to c} g(x).$

(iii) $\lim_{x \to c}(f(x)g(x)) = \left(\lim_{x \to c} f(x)\right)\left(\lim_{x \to c} g(x)\right).$

(iv) *If* $\lim_{x \to c} g(x) \neq 0$, *and* $g(x) \neq 0$ *for all* $x \in S \setminus \{c\}$, *then*

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \frac{\lim_{x \to c} f(x)}{\lim_{x \to c} g(x)}.$$

4.2 Continuity

Definition 4.2.1

f is continuous at c if $\lim_{x \to c} f(x) = f(c)$

Definition 4.2.2

Let $S \subset \mathbb{R}$, $c \in S$, and let $f : S \to \mathbb{R}$ be a function. We say that f is *continuous at c* if for every $\varepsilon > 0$ there is a $\delta > 0$ such that whenever $x \in S$ and $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.
When $f : S \to \mathbb{R}$ is continuous at all $c \in S$, then we simply say f is a *continuous function*.

Definition 4.2.3

Let x be in a neighborhood of c. Set $\Delta x = x - c$. Thus $f(x) = f(c + \Delta x)$.

$$x \to c \Leftrightarrow \Delta x \to 0$$

$f(x)$ is continuous at c if $\lim_{\Delta x \to 0} f(c + \Delta x) = f(c)$ or $\lim_{\Delta x \to 0} f(c + \Delta x) - f(c) = 0$

Definition 4.2.4

f is continuous at c if and only if for every sequence $\{x_n\}$ where $x_n \in S$ and $\lim x_n = c$, the sequence $\{f(x_n)\}$ converges to $f(c)$.

Definition 4.2.5

A function f is continuous on a set S, if continuous $\forall c \in S$.

Example.

Prove that $f(x) = sinx$ is continuous $\forall x$

Proof. Pick arbitary c. We prove $\lim_{\Delta x \to 0} sin(c + \Delta x) - sin(c) = 0$.

$$\lim_{\Delta x \to 0}(sin(c + \Delta x) - sin(c)) = \lim_{\Delta x \to 0}(2sin(\frac{\Delta x}{2})cos(\frac{\Delta x + 2c}{2}))$$

$$|2sin(\frac{\Delta x}{2})cos(\frac{\Delta x + 2c}{2})| \leq |\Delta x|, \text{ since } |cosx| \leq 1, |sinx| \leq |x|$$

Then,

$$-|\Delta x| \leq 2sin(\frac{\Delta x}{2})cos(\frac{\Delta x + 2c}{2}) \leq |\Delta x|$$

Thus, as $\Delta x \to 0, |\Delta x| \to 0$ and by the squeeze lemma, $2sin(\frac{\Delta x}{2})cos(\frac{\Delta x + 2c}{2}) \to 0$

□

Example.

$$f(x) = \frac{|x-5|}{x-5}$$

Is it continuous at $x = 5$.

No. It is not defined at $x = 5$.

Example.

$f : (0, \infty) \to \mathbb{R}$ defined by $f(x) := 1/x$ is continuous.

Proof. Fix $c \in (0, \infty)$. Let $\{x_n\}$ be a sequence in $(0, \infty)$ such that $\lim x_n = c$. Then we know that

$$f(c) = \frac{1}{c} = \frac{1}{\lim x_n} = \lim_{n \to \infty} \frac{1}{x_n} = \lim_{n \to \infty} f(x_n).$$

Thus f is continuous at c. As f is continuous at all $c \in (0, \infty)$, f is continuous. □

Proposition 4.2.6

Let $f : \mathbb{R} \to \mathbb{R}$ be a polynomial. That is

$$f(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0,$$

for some constants $a_0, a_1, \ldots, a_d$. Then f is continuous.

Proof. Fix $c \in \mathbb{R}$. Let $\{x_n\}$ be a sequence such that $\lim x_n = c$. Then

$$\begin{aligned} f(c) &= a_d c^d + a_{d-1}c^{d-1} + \cdots + a_1 c + a_0 \\ &= a_d(\lim x_n)^d + a_{d-1}(\lim x_n)^{d-1} + \cdots + a_1(\lim x_n) + a_0 \\ &= \lim_{n\to\infty} \left(a_d x_n^d + a_{d-1}x_n^{d-1} + \cdots + a_1 x_n + a_0\right) = \lim_{n\to\infty} f(x_n). \end{aligned}$$

Thus f is continuous at c. As f is continuous at all $c \in \mathbb{R}$, f is continuous. □

Proposition 4.2.7

Let $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ be functions continuous at $c \in S$.

(i) *The function $h : S \to \mathbb{R}$ defined by $h(x) := f(x) + g(x)$ is continuous at c.*

(ii) *The function $h : S \to \mathbb{R}$ defined by $h(x) := f(x) - g(x)$ is continuous at c.*

(iii) *The function $h : S \to \mathbb{R}$ defined by $h(x) := f(x)g(x)$ is continuous at c.*

(iv) *If $g(x) \neq 0$ for all $x \in S$, the function $h : S \to \mathbb{R}$ defined by $h(x) := \frac{f(x)}{g(x)}$ is continuous at c.*

Proposition 4.2.8

Let $A, B \subset \mathbb{R}$ and $f : B \to \mathbb{R}$ and $g : A \to B$ be functions. If g is continuous at $c \in A$ and f is continuous at $g(c)$, then $f \circ g : A \to \mathbb{R}$ is continuous at c.

4.3 Three Big Results

Theorem 4.3.1

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f is bounded.

Proof. Let us prove this claim by contrapositive. Suppose f is not bounded, then for each $n \in \mathbb{N}$, there is an $x_n \in [a, b]$, such that

$$|f(x_n)| \geq n.$$

Now $\{x_n\}$ is a bounded sequence as $a \leq x_n \leq b$. By the Bolzano-Weierstrass theorem, there is a convergent subsequence $\{x_{n_i}\}$. Let $x := \lim x_{n_i}$. Since $a \leq x_{n_i} \leq b$ for all i, then $a \leq x \leq b$. The limit $\lim f(x_{n_i})$ does not exist as the sequence is not bounded as $|f(x_{n_i})| \geq n_i \geq i$. On the other hand $f(x)$ is a finite number and

$$f(x) = f\left(\lim_{i\to\infty} x_{n_i}\right).$$

Thus f is not continuous at x. □

Theorem 4.3.2: Extreme Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f achieves both an absolute minimum and an absolute maximum on $[a, b]$.

Proof. Since f is continuous, it is bounded. $\exists w = sup\{f(x) : x \in [a, b]\}$, by the least upper bound property.

WTS $\exists c \in [a, b], f(c) = w$.

$\forall n \in \mathbb{N}, \exists x_n \in [a, b]$ such that $w - \frac{1}{n} < f(x_n) < w$.

By B-W, there exists a converging subsequence $(x_{n_i})_{i=1}^{\infty}$.

Say

$$\lim_{i\to\infty} x_{n_i} = c, \quad \text{with}$$

$$w - \frac{1}{n_i} < f(x_{n_i}) \leq w$$

By the squeeze theorem,

$$f(x_{n_i}) \to w.$$

Since f is continuous,

$$f(c) = \underline{f\left(\lim_{i\to\infty} x_{n_i}\right) = \lim_{i\to\infty} f(x_{n_i})} = w.$$

Proof for minimum along same (flipped lines). □

Theorem 4.3.3: Bolzano's intermediate value theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Suppose $f(a) < 0$ and $f(b) > 0$. Then there exists a number $c \in (a, b)$ such that $f(c) = 0$.

Proof. We define two sequences $\{a_n\}$ and $\{b_n\}$ inductively:

(i) Let $a_1 := a$ and $b_1 := b$.

(ii) If $f\left(\frac{a_n+b_n}{2}\right) \geq 0$, let $a_{n+1} := a_n$ and $b_{n+1} := \frac{a_n+b_n}{2}$.

(iii) If $f\left(\frac{a_n+b_n}{2}\right) < 0$, let $a_{n+1} := \frac{a_n+b_n}{2}$ and $b_{n+1} := b_n$.

From the definition of the two sequences it is obvious that if $a_n < b_n$, then $a_{n+1} < b_{n+1}$. Thus by <u>induction</u> $a_n < b_n$ for all n. Furthermore, $a_n \leq a_{n+1}$ and $b_n \geq b_{n+1}$ for all n, that is, the sequences are monotone. As $a_n < b_n \leq b_1 = b$ and $b_n > a_n \geq a_1 = a$ for all n, the sequences are also bounded. Therefore, the sequences converge. Let $c := \lim a_n$ and $d := \lim b_n$. We now want to show that $c = d$. We notice

$$b_{n+1} - a_{n+1} = \frac{b_n - a_n}{2}.$$

By <u>induction</u> we see that

$$b_n - a_n = \frac{b_1 - a_1}{2^{n-1}} = 2^{1-n}(b - a).$$

As $2^{1-n}(b - a)$ converges to zero, we take the limit as n goes to infinity to get

$$d - c = \lim_{n\to\infty}(b_n - a_n) = \lim_{n\to\infty} 2^{1-n}(b - a) = 0.$$

In other words $d = c$.
By construction, for all n we have

$$f(a_n) < 0 \qquad \text{and} \qquad f(b_n) \geq 0.$$

We use the fact that $\lim a_n = \lim b_n = c$ and the continuity of f to take limits in those inequalities to get

$$f(c) = \lim f(a_n) \leq 0 \qquad \text{and} \qquad f(c) = \lim f(b_n) \geq 0.$$

As $f(c) \geq 0$ and $f(c) \leq 0$, we conclude $f(c) = 0$. Obviously, $a < c < b$.

□

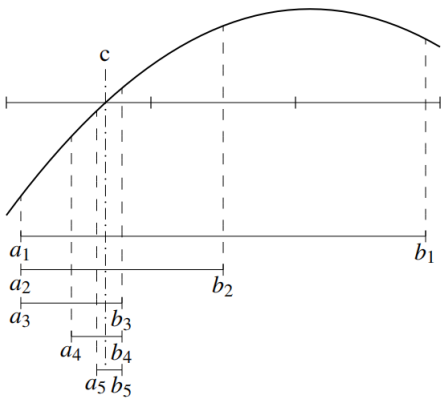


Figure 4.1: Bisection method

Theorem 4.3.4: IVT

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Suppose there exists a y such that $f(a) < y < f(b)$ or $f(a) > y > f(b)$. Then there exists a $c \in (a, b)$ such that $f(c) = y$.

Proof. If $f(a) < y < f(b)$, then define $g(x) := f(x) - y$. Then we see that $g(a) < 0$ and $g(b) > 0$ and we can apply above theorem to g. If $g(c) = 0$, then $f(c) = y$.

Similarly if $f(a) > y > f(b)$, then define $g(x) := y - f(x)$. Then again $g(a) < 0$ and $g(b) > 0$ and we can apply above theorem. Again if $g(c) = 0$, then $f(c) = y$. □

Example.

The polynomial $f(x) := x^3 - 2x^2 + x - 1$ has a real root in $(1, 2)$.

Proof. We simply notice that $f(1) = -1$ and $f(2) = 1$. Hence there must exist a point $c \in (1, 2)$ such that $f(c) = 0$. To find a better approximation of the root we could follow the proof of IVT.

For example, next we would look at 1.5 and find that $f(1.5) = -0.625$. Therefore, there is a root of the equation in $(1.5, 2)$. Next we look at 1.75 and note that $f(1.75) \approx -0.016$. Hence there is a root of f in $(1.75, 2)$. Next we look at 1.875 and find that $f(1.875) \approx 0.44$, thus there is a root in $(1.75, 1.875)$. We follow this procedure until we gain sufficient precision. □

4.4 Uniform continuity

Definition 4.4.1

Let $S \subset \mathbb{R}$, and let $f : S \to \mathbb{R}$ be a function. Suppose for any $\varepsilon > 0$ there exists a $\delta > 0$ such that whenever $x_1, x_2 \in S$ and $|x_1 - x_2| < \delta$, then $|f(x_1) - f(x_2)| < \varepsilon$. Then we say f is *uniformly continuous.*

It is not hard to see that a uniformly continuous function must be continuous. The only difference in the definitions is that for a given $\varepsilon > 0$ we pick a $\delta > 0$ that works for all $c \in S$. That is, δ can no longer depend on c, it only depends on ε. The domain of definition of the function makes a difference now. A function that is not uniformly continuous on a larger set, may be uniformly continuous when restricted to a smaller set.

$$\text{uniform continuity: } \delta(x)$$

$$\text{continuity: } \delta(x, c)$$

Example.

> $f : [0, 9] \to \mathbb{R}, f(x) = x^2$. Prove f is uniform continuity.
> On the other hand, $f : \mathbb{R} \to \mathbb{R}$, defined by $f(x) := x^2$ is not uniformly continuous.

Proof. Given $\epsilon > 0$, choose $\delta = \frac{\epsilon}{18}$. So $\forall x_1, x_2$ with $|x_1 - x_2| < \delta = \frac{\epsilon}{18}$. We have

$$|x_1^2 - x_2^2| \leq 18|x_1 - x_2| < \epsilon$$

Suppose it is uniformly continuous, then for all $\varepsilon > 0$, there would exist a $\delta > 0$ such that if $|x_1 - x_2| < \delta$, then $|x_1^2 - x_2^2| < \varepsilon$. Take $x_1 > 0$ and let $x_2 := x_1 + \delta/2$. Write

$$\varepsilon > |x_1^2 - x_2^2| = |x_1 + x_2||x_1 - x_2| = (2x_1 + \delta/2)\delta/2 \geq \delta x_1.$$

Therefore $x_1 < \varepsilon/\delta$ for all $x_1 > 0$, which is a contradiction. □

Example.

> The function $f : (a, 1) \to \mathbb{R}$ for $a > 0$, defined by $f(x) := 1/x$ is uniformly continuous.

Proof. If $a < x_1, x_2 < 1$,

$$|f(x_1) - f(x_2)| = |1/x_1 - 1/x_2| = \left|\frac{x_2 - x_1}{x_1 x_2}\right| = \frac{|x_2 - x_1|}{x_1 x_2} \leq \frac{|x_1 - x_2|}{a^2}$$

Hence $\forall \epsilon > 0$, take $\delta = \epsilon a^2$. So $\forall x_1, x_2$ with $|x_1 - x_2| < \delta$. We have

$$|f(x_1) - f(x_2)| \leq \frac{|x_1 - x_2|}{a^2} < \epsilon$$

□

Example.

> The function $f : (0, 1) \to \mathbb{R}$, defined by $f(x) := 1/x$ is not uniformly continuous, but it is continuous.

Proof. Take $\epsilon = \frac{1}{2}$. Let $\delta > 0$ be given. Set $n \in \mathbb{N}$ s.t $\frac{1}{n} < \delta$. Take $x_1 = \frac{1}{2n}, x_2 = \frac{1}{n}$, so

$$|x_1 - x_2| = \frac{1}{2n} < \frac{1}{n} < \delta$$

But

$$|f(x_1) - f(x_2)| = |\frac{1}{x_1} - \frac{1}{x_2}| = n \geq 1 \geq \frac{1}{2} = \epsilon$$

□

Definition 4.4.2: Lipschitz continuous

Let $f : S \to \mathbb{R}$ be a function such that there exists a number K such that for all x and y in S we have

$$|f(x) - f(y)| \leq K|x - y|.$$

Then f is said to be *Lipschitz continuous**.

Proposition 4.4.3

A Lipschitz continuous function is uniformly continuous.

Proof. Let $f : S \to \mathbb{R}$ be a function and let K be a constant such that for all x, y in S we have

$$|f(x) - f(y)| \leq K|x - y|.$$

Let $\varepsilon > 0$ be given. Take $\delta := \varepsilon/K$. For any x and y in S such that $|x - y| < \delta$ we have that

$$|f(x) - f(y)| \leq K|x - y| < K\delta = K \cdot \frac{\varepsilon}{K} = \varepsilon.$$

Therefore f is uniformly continuous. □

Example.

$f(x) = x^{\frac{1}{2}}, x \geq 0$ is UV but not LC.

Theorem 4.4.4

Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then f is uniformly continuous.

Chapter 5

Derivative

5.1 The Derivative

Definition 5.1.1

Let I be an interval, let $f : I \to \mathbb{R}$ be a function, and let $c \in I$. If the limit

$$L := \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists, then we say f is *differentiable* at c, that L is the *derivative* of f at c, and write $f'(c) := L$.

Example.

Let $f(x) := x^2$ defined on the whole real line. We find that

$$\lim_{x \to c} \frac{x^2 - c^2}{x - c} = \lim_{x \to c} \frac{(x + c)(x - c)}{x - c} = \lim_{x \to c} (x + c) = 2c.$$

Therefore $f'(c) = 2c$.

Example.

The function $f(x) := |x|$ is not differentiable at the origin.

When $x > 0$, then

$$\frac{|x| - |0|}{x - 0} = \frac{x - 0}{x - 0} = 1,$$

and when $x < 0$ we have

$$\frac{|x| - |0|}{x - 0} = \frac{-x - 0}{x - 0} = -1.$$

Hence lim at $x = 0$ does not exist.

Proposition 5.1.2

Let $f : I \to \mathbb{R}$ be differentiable at $c \in I$, then it is continuous at c.

Proof. We know the limits

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c) \quad \text{and} \quad \lim_{x \to c} (x - c) = 0$$

exist. Furthermore,

$$f(x) - f(c) = \left(\frac{f(x) - f(c)}{x - c} \right) (x - c).$$

Therefore the limit of $f(x) - f(c)$ exists and

$$\lim_{x\to c}(f(x) - f(c)) = \left(\lim_{x\to c}\frac{f(x) - f(c)}{x - c}\right)\left(\lim_{x\to c}(x - c)\right) = f'(c) \cdot 0 = 0.$$

Hence $\lim_{x\to c} f(x) = f(c)$, and f is continuous at c. □

Proposition 5.1.3

Let I be an interval, let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be differentiable at $c \in I$, and let $\alpha \in \mathbb{R}$.

(i) *Define $h : I \to \mathbb{R}$ by $h(x) := \alpha f(x)$. Then h is differentiable at c and $h'(c) = \alpha f'(c)$.*

(ii) *Define $h : I \to \mathbb{R}$ by $h(x) := f(x) + g(x)$. Then h is differentiable at c and $h'(c) = f'(c) + g'(c)$.*

Proof. First, let $h(x) := \alpha f(x)$. For $x \in I$, $x \neq c$ we have

$$\frac{h(x) - h(c)}{x - c} = \frac{\alpha f(x) - \alpha f(c)}{x - c} = \alpha\frac{f(x) - f(c)}{x - c}.$$

We get

$$\lim_{x\to c}\frac{h(x) - h(c)}{x - c} = \alpha\lim_{x\to c}\frac{f(x) - f(c)}{x - c}.$$

Therefore h is differentiable at c, and the derivative is computed as given.

Next, define $h(x) := f(x) + g(x)$. For $x \in I$, $x \neq c$ we have

$$\frac{h(x) - h(c)}{x - c} = \frac{(f(x) + g(x)) - (f(c) + g(c))}{x - c} = \frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c}.$$

We get

$$\lim_{x\to c}\frac{h(x) - h(c)}{x - c} = \lim_{x\to c}\frac{f(x) - f(c)}{x - c} + \lim_{x\to c}\frac{g(x) - g(c)}{x - c}.$$

Therefore h is differentiable at c and the derivative is computed as given. □

Proposition 5.1.4

(Product rule). *Let I be an interval, let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be functions differentiable at c. If $h : I \to \mathbb{R}$ is defined by*

$$h(x) := f(x)g(x),$$

then h is differentiable at c and

$$h'(c) = f(c)g'(c) + f'(c)g(c).$$

Proof.

$$Q_{h,c}^{(x)} = \frac{f(x)g(x) - f(c)g(c)}{x - c} = \frac{f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)}{x - c}$$
$$= f(x)Q_{g,c}^{(x)} + g(c)Q_{f,c}^{(x)}$$

We get

$$\lim_{x\to c} Q_{h,c}^{(x)} = \lim_{x\to c} f(x)Q_{g,c}^{(x)} + \lim_{x\to c} g(c)Q_{f,c}^{(x)} = f(c)\lim_{x\to c} Q_{g,c}^{(x)} + g(c)\lim_{x\to c} Q_{f,c}^{(x)}.$$

Which is

$$h'(c) = f(c)g'(c) + f'(c)g(c).$$

□

Proposition 5.1.5

(Chain Rule). *Let I_1, I_2 be intervals, let $g : I_1 \to I_2$ be differentiable at $c \in I_1$, and $f : I_2 \to \mathbb{R}$ be differentiable at $g(c)$. If $h : I_1 \to \mathbb{R}$ is defined by*

$$h(x) := (f \circ g)(x) = f(g(x)),$$

then h is differentiable at c and

$$h'(c) = f'(g(c))g'(c).$$

Proof. Let $d := g(c)$. Define $u : I_2 \to \mathbb{R}$ and $v : I_1 \to \mathbb{R}$ by

$$u(y) := \begin{cases} \frac{f(y)-f(d)}{y-d} & \text{if } y \neq d, \\ f'(d) & \text{if } y = d, \end{cases} \qquad v(x) := \begin{cases} \frac{g(x)-g(c)}{x-c} & \text{if } x \neq c, \\ g'(c) & \text{if } x = c. \end{cases}$$

We note that

$$f(y) - f(d) = u(y)(y - d) \quad \text{and} \quad g(x) - g(c) = v(x)(x - c).$$

We plug in to obtain

$$h(x) - h(c) = f(g(x)) - f(g(c)) = u(g(x))(g(x) - g(c)) = u(g(x))(v(x)(x - c)).$$

Therefore,

$$\frac{h(x) - h(c)}{x - c} = u(g(x))v(x).$$

We compute the limits $\lim_{y\to d} u(y) = f'(d) = f'(g(c))$ and $\lim_{x\to c} v(x) = g'(c)$. That is, the functions u and v are continuous at $d = g(c)$ and c respectively. Furthermore, the function g is continuous at c. Hence the limit of the right-hand side of (4.1) as $x \to c$ exists and is equal to $f'(g(c))g'(c)$. Thus h is differentiable at c and the limit is $f'(g(c))g'(c)$.

□

Proposition 5.1.6

(Quotient Rule). *Let I be an interval, let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be differentiable at c and $g(x) \neq 0$ for all $x \in I$. If $h : I \to \mathbb{R}$ is defined by*

$$h(x) := \frac{f(x)}{g(x)},$$

then h is differentiable at c and

$$h'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Proof. Outline:

$$\begin{aligned} h(x) &= \frac{f(x)}{g(x)} = f(x)u(g(x)) \\ h'(c) &= f'(c)u(g(c)) + f(c)u'(g(c)) \\ &= \frac{f'(c)}{g(c)} + f(c)\frac{-1}{g(c)^2}g'(c) \\ &= \frac{f'(c)g(c)}{g(c)^2} - \frac{f(c)g'(c)}{g(c)^2} \end{aligned}$$

□

Example.

$f(x) = sinx$. Prove $f'(x) = cosx$

Proof. Hint: $sin\alpha - sin\beta = 2sin(\frac{\alpha-\beta}{2})cos(\frac{\alpha+\beta}{2})$

$$f'(c) = \lim_{\Delta x \to 0} \frac{f(c+\Delta x) - f(c)}{\Delta x} = \lim_{\Delta x \to 0} \frac{sin(c+\Delta x) - sinc}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{2}{\Delta x} sin(\frac{\Delta x}{2}) cos(\frac{2c+\Delta x}{2})$$

$$= \lim_{\Delta x \to 0} \frac{sin(\frac{\Delta x}{2})}{\frac{\Delta x}{2}} \lim_{\Delta x \to 0} cos(\frac{2c+\Delta x}{2})$$

$$= cos(c) \lim_{\Delta x \to 0} \frac{sin(\frac{\Delta x}{2})}{\frac{\Delta x}{2}}$$

Claim: $\lim_{x\to 0} \frac{sinx}{x} = 1$. Prove by Squeeze theorem. Then

$$f'(c) = cosc$$

□

5.2 Mean value theorem

Definition 5.2.1

Let $S \subset \mathbb{R}$ be a set and let $f : S \to \mathbb{R}$ be a function. The function f is said to have a *relative (local) maximum* at $c \in S$ if there exists a $\delta > 0$ such that for all $x \in S$ such that $|x - c| < \delta$ we have $f(x) \leq f(c)$. The definition of *relative (local) minimum* is analogous.

Lemma 5.2.2

Let $f : [a, b] \to \mathbb{R}$ be a function differentiable at $c \in (a, b)$, and c is a relative (local) minimum or a relative (local) maximum of f. Then $f'(c) = 0$.

Proof. We prove the statement for a maximum. For a minimum the statement follows by considering the function $-f$.

Let c be a relative maximum of f. In particular, as long as $|x - c| < \delta$ we have $f(x) - f(c) \leq 0$. Then we look at the difference quotient. If $x > c$ we note that

$$\frac{f(x) - f(c)}{x - c} \leq 0,$$

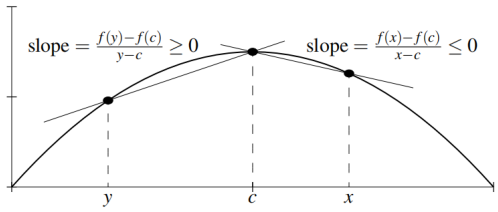
and if $y < c$ we have

$$\frac{f(y) - f(c)}{y - c} \geq 0.$$

We now take sequences $\{x_n\}$ and $\{y_n\}$ such that $x_n > c$ and $y_n < c$ for all $n \in \mathbb{N}$, and such that $\lim x_n = \lim y_n = c$. Since f is differentiable at c we know

$$0 \geq \lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} = f'(c) = \lim_{n\to\infty} \frac{f(y_n) - f(c)}{y_n - c} \geq 0.$$

□



Theorem 5.2.3: Rolle's Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function differentiable on (a, b) such that $f(a) = f(b)$. Then there exists a $c \in (a, b)$ such that $f'(c) = 0$.

Proof. As f is continuous on $[a, b]$ it attains an absolute minimum and an absolute maximum in $[a, b]$. We wish to apply **Theorem** and so we need a minimum or maximum at some $c \in (a, b)$. Write $K := f(a) = f(b)$. If there exists an x such that $f(x) > K$, then the absolute maximum is bigger than K and hence occurs at $c \in (a, b)$, and therefore we get $f'(c) = 0$. On the other hand if there exists an x such that $f(x) < K$, then the absolute minimum occurs at some $c \in (a, b)$ and we have that $f'(c) = 0$. If there is no x such that $f(x) > K$ or $f(x) < K$, then we have that $f(x) = K$ for all x and then $f'(x) = 0$ for all $x \in [a, b]$, so any c will work.

□

Theorem 5.2.4: Mean Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous function differentiable on (a, b). Then there exists a point $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. The theorem follows from *Rolle's theorem*. Define the function $g : [a, b] \to \mathbb{R}$ by

$$g(x) := f(x) - f(b) + (f(b) - f(a))\frac{b - x}{b - a}.$$

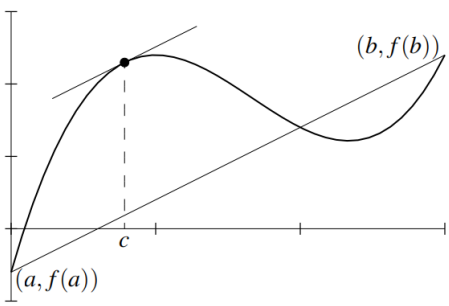
The function g is differentiable on (a, b), continuous on $[a, b]$, such that $g(a) = 0$ and $g(b) = 0$. Thus there exists $c \in (a, b)$ such that $g'(c) = 0$.

$$0 = g'(c) = f'(c) + (f(b) - f(a))\left(-\frac{1}{b - a}\right).$$

Or in other words

$$f'(c)(b - a) = f(b) - f(a).$$

□



Example.

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) := \begin{cases} (x \sin(1/x))^2 & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

We claim that f is differentiable everywhere, but $f' : \mathbb{R} \to \mathbb{R}$ is not continuous at the origin. Furthermore, f has a minimum at 0, but the derivative changes sign infinitely often near the origin.